Time: 30 minutes Max marks: 15

Instructions:
- Do not plagiarize. Do not assist your classmates in plagiarism.
- Define all your variables properly, and show your full solution to get full credit.
- Attempt all questions that you can.
- In the unlikely case a question is not clear, discuss it with an invigilating TA/TF. Please ensure that you clearly include any assumptions you make, even after clarification from the invigilator.

1. (5 points) **Logistic Regression and Sigmoid Function**.

   (a) (3 points) Show that the derivative of the sigmoid function ($\sigma(z) = \frac{1}{1+e^{-z}}$) can be written as a function of $\sigma(z)$ itself, i.e., $\frac{\partial \sigma(z)}{\partial z} = g(\sigma(z))$. Find out what is $g(\sigma(z))$.

   (b) (2 points) What is the inverse of the sigmoid function? Given the probability $p = \sigma(z)$, compute the logit $z = \boldsymbol{w}^T \boldsymbol{x}$.

Solution:

(a) **Derivative of the Sigmoid Function**

The sigmoid function is defined as:

$$\sigma(z) = \frac{1}{1+e^{-z}} = (1+e^{-z})^{-1}$$

We compute the derivative with respect to z using the chain rule:

$$\frac{\partial \sigma(z)}{\partial z} = -(1+e^{-z})^{-2} \cdot \frac{\partial}{\partial z}(1+e^{-z})$$

$$\frac{\partial \sigma(z)}{\partial z} = -(1+e^{-z})^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1+e^{-z})^2}$$

We can rewrite this expression by splitting the denominator:

$$\frac{\partial \sigma(z)}{\partial z} = \left(\frac{1}{1+e^{-z}}\right) \cdot \left(\frac{e^{-z}}{1+e^{-z}}\right)$$

Notice that:

$$1 - \sigma(z) = 1 - \frac{1}{1+e^{-z}} = \frac{1+e^{-z}-1}{1+e^{-z}} = \frac{e^{-z}}{1+e^{-z}}$$

Substituting $\sigma(z)$ and $1-\sigma(z)$ back into the derivative formula yields:

$$\frac{\partial \sigma(z)}{\partial z} = \sigma(z)(1-\sigma(z))$$

Thus, the function $g(\sigma(z))$ is given by:

$$g(\sigma(z)) = \sigma(z)(1-\sigma(z))$$

(b) **Inverse of the Sigmoid Function (Logit)**

To find the inverse function, we set $p = \sigma(z)$ and solve for z:

$$p = \frac{1}{1+e^{-z}}$$

Taking the reciprocal of both sides:

$$1 + e^{-z} = \frac{1}{p}$$

Subtracting 1 from both sides:

$$e^{-z} = \frac{1}{p} - 1 = \frac{1-p}{p}$$

Taking the reciprocal again to eliminate the negative exponent:

$$e^{z} = \frac{p}{1-p}$$

Taking the natural logarithm of both sides gives the logit function:

$$z = \ln\left(\frac{p}{1-p}\right)$$

Therefore, the inverse of the sigmoid function is the logit function $\text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$. Given $p = \sigma(z)$, the expression for the logit is:

$$z = \boldsymbol{w}^T \boldsymbol{x} = \ln\left(\frac{p}{1-p}\right)$$

Rubric / Marks Distribution

Marking Criterion	**Marks**
(a) Derivative of Sigmoid Function – 3 marks	
Correctly writes the sigmoid function and applies the chain rule	1
Correctly derives $\frac{e^{-z}}{(1+e^{-z})^2}$ and rewrites it in terms of sigmoid components	1
Correctly obtains $g(\sigma(z)) = \sigma(z)(1-\sigma(z))$	1
(b) Inverse Sigmoid / Logit – 2 marks	
Correct algebraic manipulation starting from $p = \sigma(z)$	1
Correctly obtains $z = \log\left(\frac{p}{1-p}\right) = \mathbf{w}^T\mathbf{x}$	1
Total	**5**

2. (5 points) **Maximum Likelihood Estimation**.
Harvey Dent (from Christopher Nolan's 2008 film "The Dark Knight") gets hold of Jai's coin from the 1975 film "Sholay". Over the 33 years, the coin has grown a conscience, and has become less biased. To estimate how biased is the coin, Thakur (also from Sholay) asks Ramlal to flip the coin a hundred times (remember, Thakur doesn't have arms!) and count the number of heads, and it turns out to be 70! Thakur concludes that the probability of heads ($P(X = Heads) = p = 0.7$), but asks your opinion. Convince Thakur and yourself that Thakur's estimate is the Maximum Likelihood Estimate under the modeling assumption that the coin is a Bernoulli random variable with parameter p. Write the Bernoulli distribution PMF, the likelihood function, and maximize it w.r.t. p to show that the $\widehat{p}_{MLE}$ is indeed the normalized counts of the number of heads appearing in the series of experiments.

Solution:

1. Maximum Likelihood Estimation Problem Modeling

Let X_i be a random variable representing the outcome of the i-th coin flip, where $i \in \{1, 2, \ldots, n\}$ and $n = 100$. We assume each flip is independent and identically distributed (i.i.d.) following a Bernoulli distribution with parameter p, where $p = P(X_i = 1)$ is the probability of getting a Head. Let $X_i = 1$ denote Heads and $X_i = 0$ denote Tails. The Probability Mass Function (PMF) for a single Bernoulli trial is given by:

$$P(X_i = x_i) = p^{x_i}(1-p)^{1-x_i} \quad \text{for } x_i \in \{0, 1\}$$

2. The Likelihood Function

Given a dataset of observed outcomes $\mathcal{D} = \{x_1, x_2, \ldots, x_n\}$, the joint probability (Likelihood function $L(p)$) of observing this specific sequence due to independence is the product of individual PMFs:

$$L(p) = \prod_{i=1}^{n} P(X_i = x_i) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i}$$

Let $k = \sum_{i=1}^{n} x_i$ be the total number of Heads observed (here, $k = 70$). The number of Tails is consequently $n - k = 30$. Simplifying the likelihood expression gives:

$$L(p) = p^{\sum_{i=1}^{n} x_i}(1-p)^{n-\sum_{i=1}^{n} x_i} = p^k(1-p)^{n-k}$$

3. The Log-Likelihood Function

To maximize the likelihood with respect to p, it is mathematically simpler to maximize its natural logarithm (the log-likelihood, $\ln L(p)$), since the logarithm is a strictly increasing monotonic transformation:

$$\ln L(p) = \ln \left[p^k(1-p)^{n-k}\right]$$

Applying logarithmic identities yields:

$$\ln L(p) = k \ln(p) + (n-k)\ln(1-p)$$

4. Maximizing the Log-Likelihood

To find the Maximum Likelihood Estimate ($\widehat{p}_{\text{MLE}}$), we take the first derivative of the log-likelihood function with respect to p and set it to zero:

$$\frac{d}{dp}\left[\ln L(p)\right] = 0$$

$$\frac{k}{p} - \frac{n-k}{1-p} = 0$$

Solving for p:

$$\frac{k}{p} = \frac{n-k}{1-p}$$

$$k(1-p) = p(n-k)$$

$$k - kp = np - kp$$

$$k = np \implies \widehat{p}_{\text{MLE}} = \frac{k}{n}$$

5. Verification of Maximum

To guarantee that $\widehat{p}_{\text{MLE}}$ corresponds to a maximum (and not a minimum), we compute the second derivative of the log-likelihood function:

$$\frac{d^2}{dp^2}\left[\ln L(p)\right] = -\frac{k}{p^2} - \frac{n-k}{(1-p)^2}$$

Since $k > 0$, $n - k > 0$, and the denominators are squared, both terms are strictly negative for any valid probability $p \in (0, 1)$. Thus, $\frac{d^2}{dp^2}\left[\ln L(p)\right] < 0$, verifying a strict concave maximum.

6. The MLE Estimate

The maximum likelihood estimator $\widehat{p}_{\text{MLE}} = \frac{k}{n}$ proves that the optimal parameter is exactly the normalized count of heads. Plugging the actual counts from Ramlal's experiments, we get $k = 70$ heads out of $n = 100$ flips:

$$\widehat{p}_{\text{MLE}} = \frac{70}{100} = 0.7$$

Thakur's intuition is mathematically verified. The estimate $p = 0.7$ is strictly the *Maximum Likelihood Estimate* under these Bernoulli modeling assumptions.

Rubric / Marks Distribution

Marking Criterion	Marks
Correct Bernoulli PMF: $P(X_i = x_i) = p^{x_i}(1-p)^{1-x_i}$	1
Correctly formulates the likelihood function and simplifies it to $L(p) = p^k(1-p)^{n-k}$	1
Correctly derives the log-likelihood $\log L(p) = k \log p + (n - k)\log(1-p)$	1
Correct differentiation and solution of the first-order condition, giving $\widehat{p}_{MLE} = \frac{k}{n}$	1
Verifies the maximum using the second derivative and substitutes $k = 70,\ n = 100$ to obtain $\widehat{p}_{MLE} = 0.7$	1
Total	**5**

3. (5 points) **Linear Regression & Regularization**.
Dent stealing Jai's unfair coin has inspired the Joker to sabotage the crime rate prediction model of Gotham's Data Analytics Team, which is now led by Thakur upon Commissioner Gordon's recommendation. Thakur is using a linear regression model to predict the crime rate ($y \in \mathbb{R}$), but the data matrix $\mathbf{X} \in \mathbb{R}^{d \times N}$, with N instances of d-dimensional input data, has been partly corrupted by the Joker and made to be strictly rank-deficient, i.e., $(\text{rank}(\mathbf{X}) < \min(d, N))$. Unaware of this data corruption, and running out of patience, Thakur is not armed with any iterative approach, so he insists that you to find the closed-form solution for the prediction model parameters. Using matrix and L_2 norm notation, you have to write down the loss function that you would optimize to solve the linear regression problem, and find the optimal parameters in closed form. Derive the solution for the impatient and irritable Thakur, and recommend the solution only if you can compute it, otherwise suggest an appropriate alternate solution. {*Hint*: Meanwhile, in Ramgarh, Veeru is trying to look away to avoid seeing Basanti dance in front of the dogs, and spots Samba sitting afar on a ridge, regressing into an oblivious role in Gabbar's gang.}

Solution:

1. Problem Identification & Objective

Thakur is attempting to find a closed-form solution to a linear regression problem where the data matrix $\mathbf{X} \in \mathbb{R}^{d \times N}$ is strictly rank-deficient, i.e., $\text{rank}(\mathbf{X}) < \min(d, N)$.

Let the linear prediction model be:

$$\hat{\mathbf{y}} = \mathbf{X}^T \mathbf{w}$$

where $\mathbf{w} \in \mathbb{R}^d$ is the parameter vector and $\mathbf{y} \in \mathbb{R}^N$ is the target crime rate vector.

In standard ordinary least squares (OLS) regression, the loss function is the sum of squared errors:

$$L_{\text{OLS}}(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}^T \mathbf{w}\|_2^2$$

The standard closed-form solution is found by solving the normal equations:

$$(\mathbf{X}\mathbf{X}^T)\mathbf{w} = \mathbf{X}\mathbf{y}$$

However, because $\mathbf{X}$ is rank-deficient ($\text{rank}(\mathbf{X}) < d$), the matrix $\mathbf{X}\mathbf{X}^T \in \mathbb{R}^{d \times d}$ has a rank less than d and is therefore **singular (not invertible)**. Thus, standard OLS has no unique closed-form solution.

To overcome this and provide a guaranteed computable closed-form solution for the impatient Thakur, we must introduce **L_2 regularization (Ridge Regression)**.

2. Loss Function with L_2 Regularization

The regularized loss function using matrix and L_2 norm notation is:

$$L(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}^T \mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_2^2$$

where $\lambda > 0$ is the regularization parameter.

3. Derivation of the Closed-Form Solution

We expand the loss function using matrix properties:

$$L(\mathbf{w}) = (\mathbf{y} - \mathbf{X}^T \mathbf{w})^T (\mathbf{y} - \mathbf{X}^T \mathbf{w}) + \lambda \mathbf{w}^T \mathbf{w}$$

$$L(\mathbf{w}) = \mathbf{y}^T \mathbf{y} - 2\mathbf{w}^T \mathbf{X}\mathbf{y} + \mathbf{w}^T \mathbf{X}\mathbf{X}^T \mathbf{w} + \lambda \mathbf{w}^T \mathbf{w}$$

To minimize $L(\mathbf{w})$, we take the gradient with respect to $\mathbf{w}$ and set it to zero:

$$\nabla_{\mathbf{w}} L(\mathbf{w}) = -2\mathbf{X}\mathbf{y} + 2\mathbf{X}\mathbf{X}^T \mathbf{w} + 2\lambda \mathbf{w} = \mathbf{0}$$

$$(\mathbf{X}\mathbf{X}^T + \lambda \mathbf{I}_d)\mathbf{w} = \mathbf{X}\mathbf{y}$$

where $\mathbf{I}_d$ is the $d \times d$ identity matrix.

Since $\mathbf{X}\mathbf{X}^T$ is symmetric and positive semi-definite, adding $\lambda \mathbf{I}_d$ (with $\lambda > 0$) ensures that $(\mathbf{X}\mathbf{X}^T + \lambda \mathbf{I}_d)$ is strictly **positive definite and invertible**, regardless of the rank of $\mathbf{X}$.

Thus, the unique optimal closed-form solution for ridge regression is:

$$\mathbf{w}^* = (\mathbf{X}\mathbf{X}^T + \lambda \mathbf{I}_d)^{-1} \mathbf{X}\mathbf{y} \tag{1}$$

The above derivation yields the closed form solution for the ordinary least squares (OLS) problem by substituting $\lambda = 0$ everywhere above, with the OLS solution being:

$$\boldsymbol{w}^*_{OLS} = (\mathbf{X}\mathbf{X}^T)^{-1}\mathbf{X}\boldsymbol{y}$$

Of course, the above is only computable if $\mathbf{X}$ is *not* rank-deficient. Since in the given settings, $\mathbf{X}$ is rank-deficient, we can only use the alternate closed form solution of ridge regression as computed in eqn. (1).

Rubric / Marks Distribution

Marking Criterion	**Marks**
Correctly identifies the OLS loss and explains that $\mathbf{X}\mathbf{X}^T$ is singular due to rank deficiency	1
Correctly introduces L_2 regularization and writes $L(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}^T\mathbf{w}\|_2^2 + \lambda\|\mathbf{w}\|_2^2$	1
Correct expansion of the loss and computation of the gradient	1
Correctly derives $(\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I}_d)\mathbf{w} = \mathbf{X}\mathbf{y}$ and explains why the matrix is invertible for $\lambda > 0$	1
Correct closed-form ridge solution $\mathbf{w}^* = (\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I}_d)^{-1}\mathbf{X}\mathbf{y}$ and correctly concludes that ridge regression is the computable alternative for the given rank-deficient $\mathbf{X}$	1
Total	**5**